

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SAAWI_dvw0")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1212)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,          100,          0.2,          40, 2000);
Distrib (Fm,           TruncNormal,              0.1,           0.2,           0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,          0.8,           0.1,           0.1, 2);
Distrib (kap,          Uniform,                  0.1,           0.9);
Distrib (kapA,         Uniform,                  0.1,           0.9);
Distrib (Ehp,          TruncNormal_cv,          100,           0.2,           40, 300);
Distrib (E_coc,        TruncNormal_cv,          30,            0.2,           1, 300);
Distrib (rOM_ClxHorse, LogUniform,              0.00001,       0.3);
Distrib (f_Slope, TruncNormal_cv, 1, 1, 0, 10);
Distrib (dv, TruncNormal_cv, 2, 0.5, 0, 10);
Distrib (w0, LogUniform, 0.000000001, 0.1);
Distrib (f_EC, TruncNormal_cv, 0.5, 0.1, 0.1, 0.9);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcaliSWI DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	kapA.1.	Ehp.1.
Result_mode	196.56500	0.6854560	0.82611000	0.899019000	0.64292300	133.06200
2.5%	175.40600	0.4269880	0.67206800	0.875993650	0.55622900	110.67455

97.5%	213.14640	0.7957250	0.90762190	0.899873000	0.72846800	193.02040
Min	87.09600	0.1686680	0.57377400	0.353873000	0.18063900	76.84800
Max	226.99200	0.7999990	0.98692000	0.900000000	0.80261500	236.19800
Result_mean	194.13107	0.6588853	0.79461172	0.893916457	0.63614841	144.29874
Result_sd	9.53307	0.1046416	0.06184647	0.007799643	0.04403934	20.77671
	E_coc.1.	rOM_ClxHorse.1.	f_Slope.1.	dv.1.	w0.1.	
Result_mode	19.945900	0.003000160	0.6081770	4.5130700	1.051230e-09	
2.5%	16.509630	0.001705380	0.4758900	2.3534650	1.051230e-09	
97.5%	28.533765	0.008725408	1.8483030	5.6460410	2.265373e-05	
Min	13.848400	0.000105423	0.3539590	1.2078200	1.040430e-09	
Max	35.834800	0.012683600	2.9258000	6.9589600	4.262110e-03	
Result_mean	21.762734	0.004395297	0.7651851	4.1671432	2.424956e-06	
Result_sd	3.067158	0.001787578	0.3389724	0.8697691	2.927530e-05	
	f_EC.1.	Sigma_W.1.				
Result_mode	0.5202050	0.12361500				
2.5%	0.4297993	0.10990595				
97.5%	0.6264230	0.14113400				
Min	0.3610670	0.09712150				
Max	0.7080920	6.00112000				
Result_mean	0.5295318	0.12477869				
Result_sd	0.0499382	0.05248016				

Number of observations, n = 277
 Number of parameters, k = 12
 Log-likelihood, LL = -11.57871
 BIC = 90.64563

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.18	1.53
E_coc.1.	1.17	1.52
Ehp.1.	1.11	1.35
f_EC.1.	1.02	1.05
f_Slope.1.	1.05	1.16
Fm.1.	1.03	1.08
kap.1.	1.08	1.20
kapA.1.	1.08	1.24
pAm.1.	1.03	1.04
r_pAm_pM.1.	1.16	1.49
rOM_ClxHorse.1.	1.01	1.01

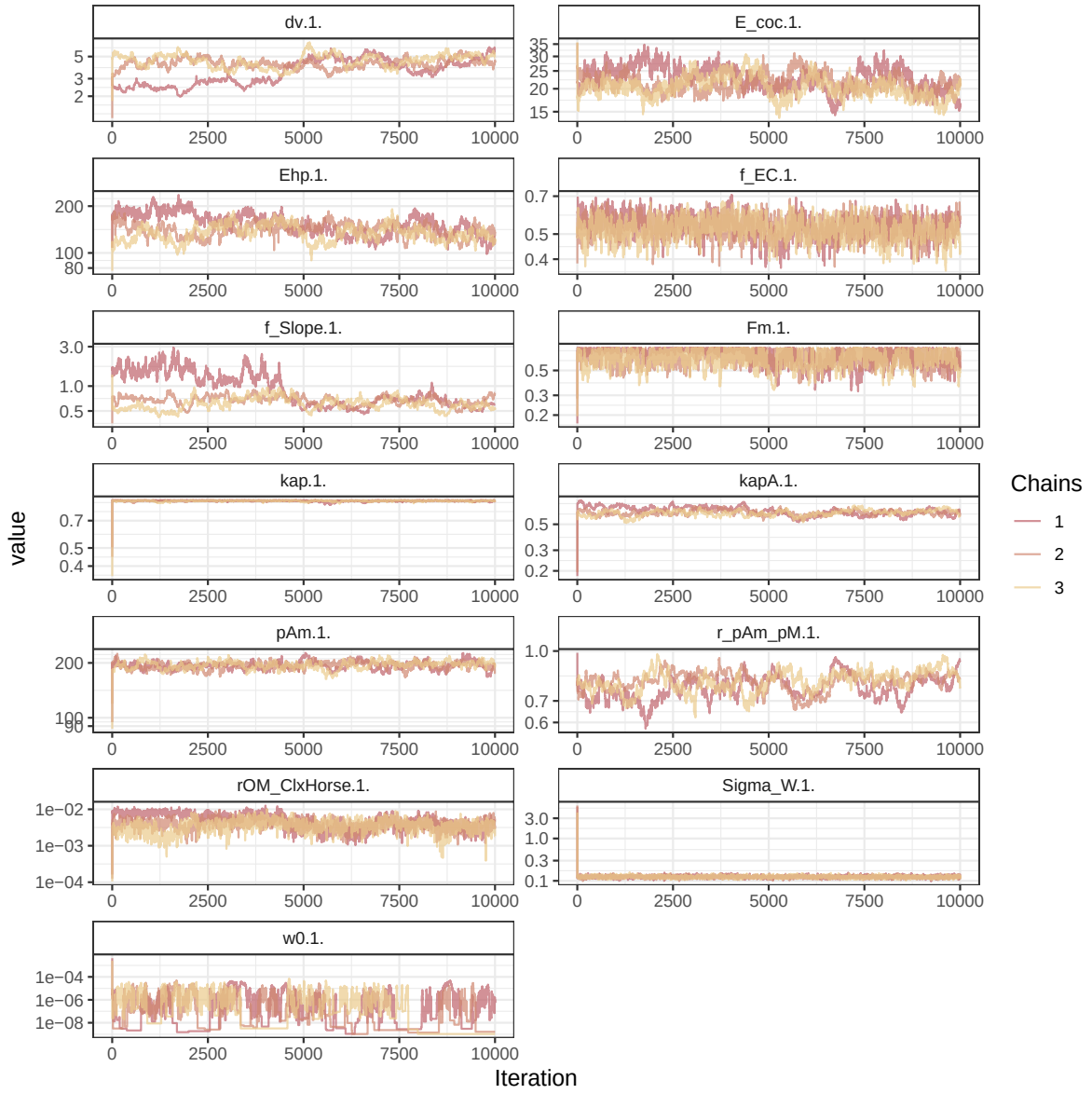


Figure 1: Parameters chains. Only the first iteration and the last r `Nb_iter_kept` are represented.

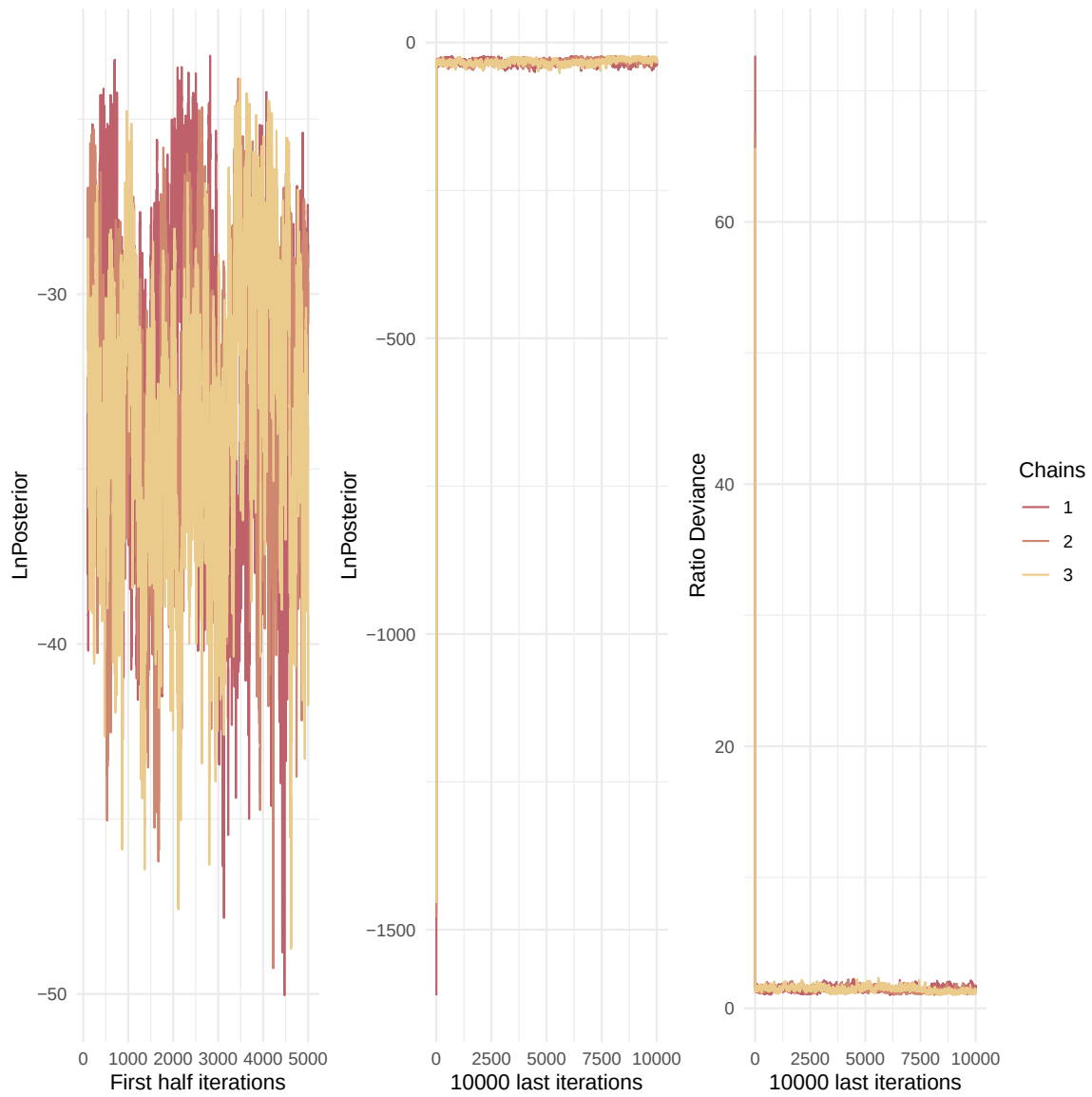


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

Sigma_W.1.	1.00	1.01
w0.1.	1.11	1.20

Multivariate psrf

1.16

```
mcmc_list_corr <- mcmc.list(  
  lapply(mcmc_list, function(x) {  
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc  
  })  
)  
  
colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))  
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

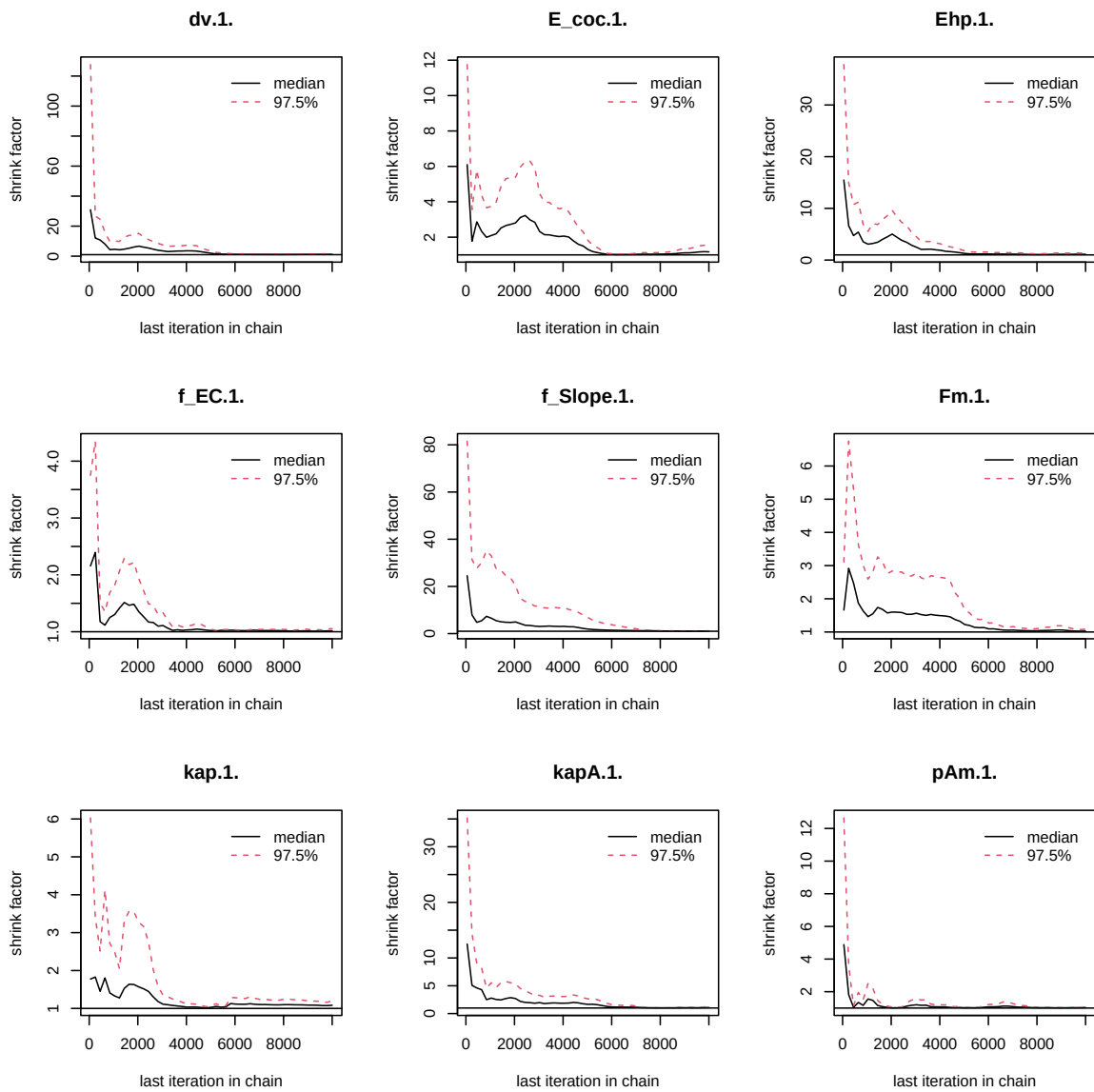


Figure 3: Chains convergence.

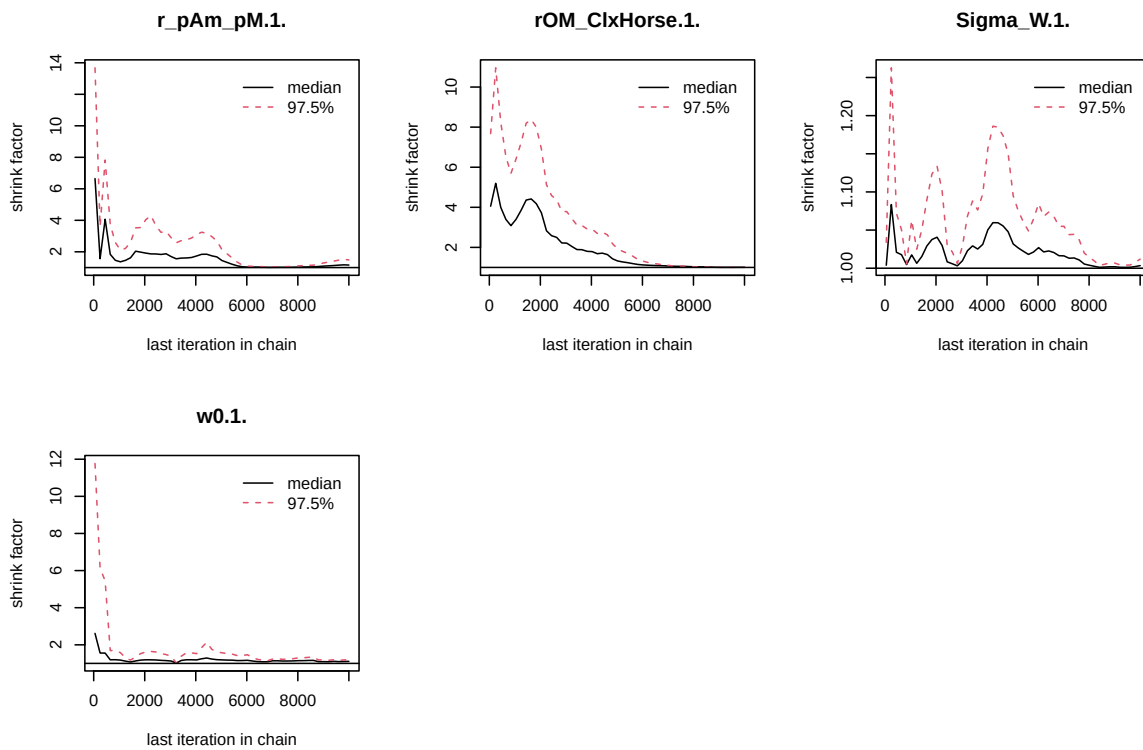


Figure 4: Chains convergence.

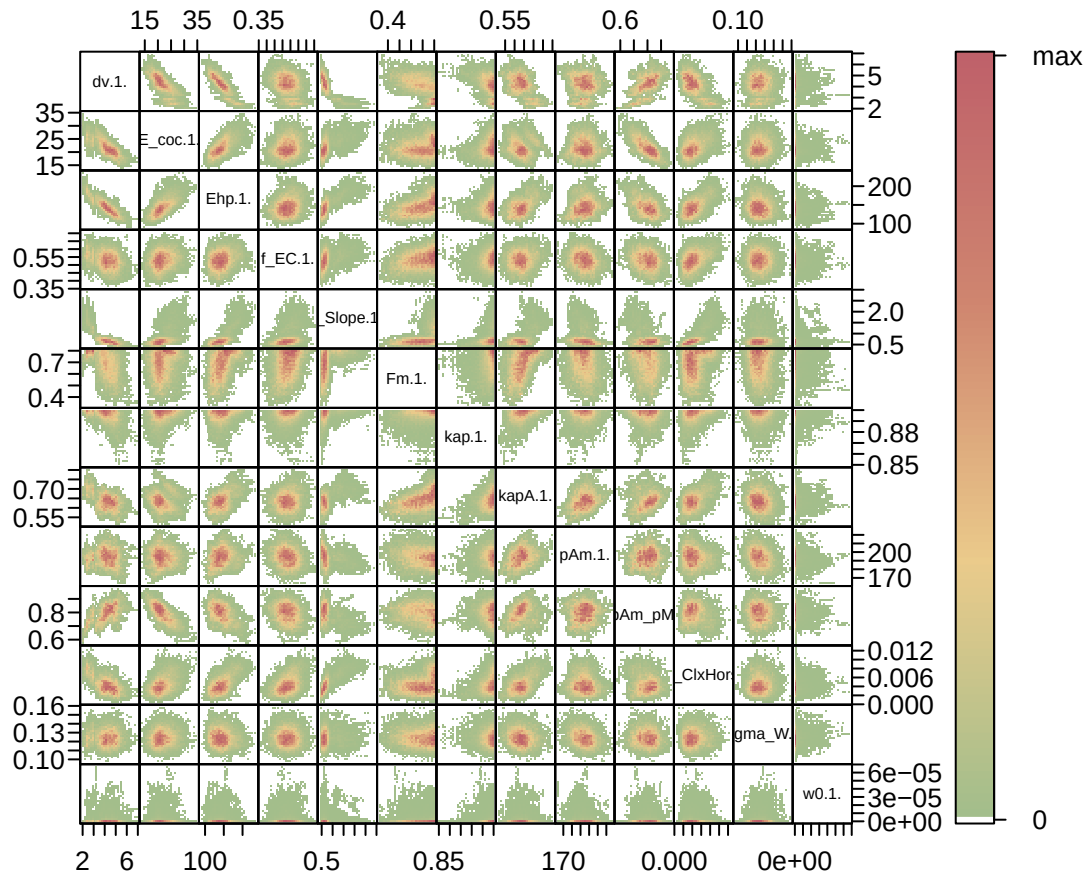


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",

```

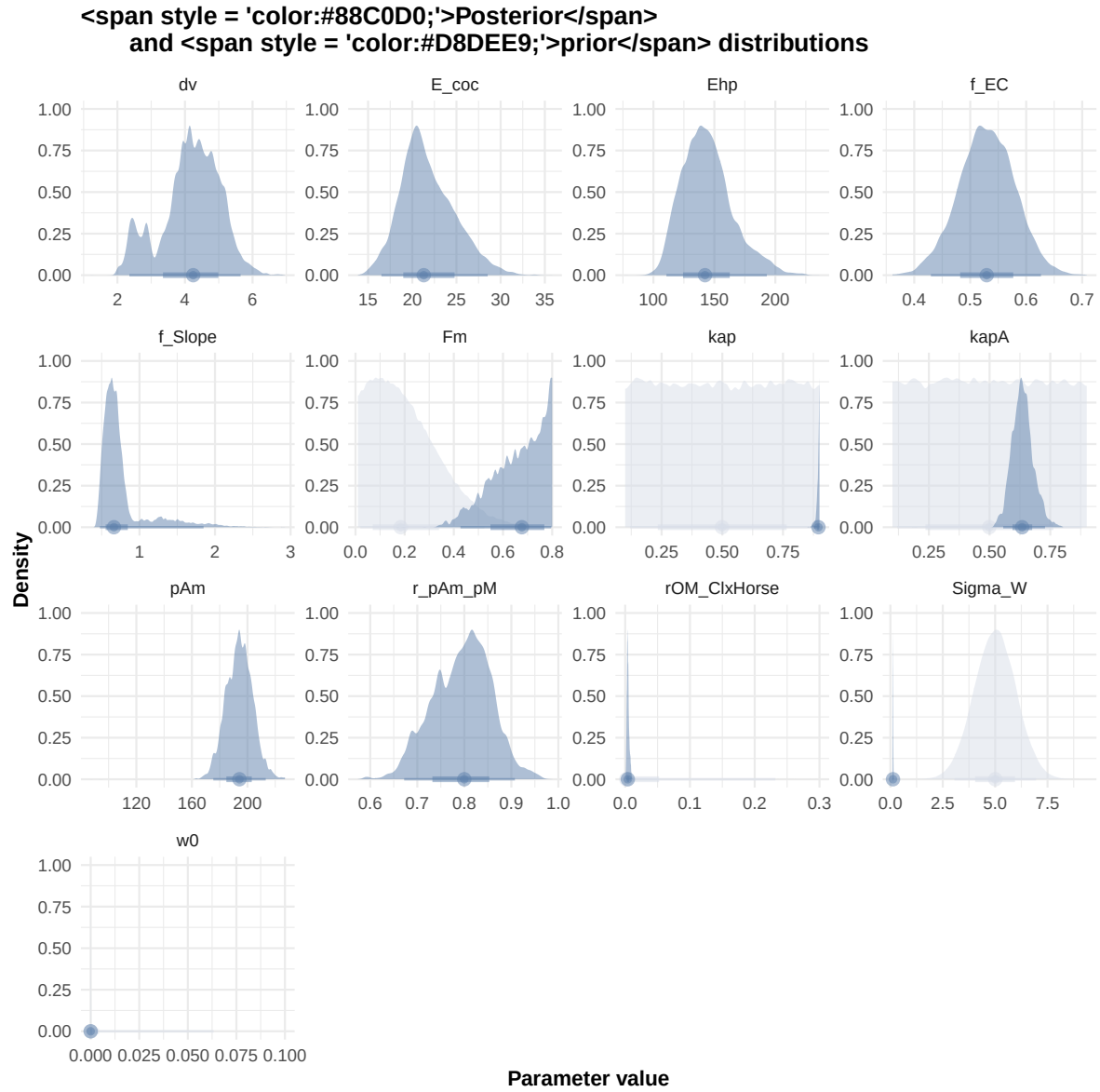


Figure 6: Distributions of the priors and the posteriors.

```
"Bart2020",  
#"Gollot2026_lufa"  
"Gollot2026_dens_D1",  
"Gollot2026_dens_D2",  
"Gollot2026_dens_D2b",  
"Gollot2026_dens_D5",  
"Gollot2026_dens_D7"  
)  
  
Adults_alone <- FALSE  
  
df_data <- f_import_data_DEB(l_Experiments, Adults_alone)  
  
text_param <- Summary_res["Result_mode", , drop = FALSE] %>%  
  t() %>%  
  as.data.frame() %>%  
  tibble::rownames_to_column("param") %>%  
  rename(val = 2) %>%  
  mutate(  
    param = gsub("\\\\.1\\.$|\\.1\\.\"", "", param),  
    line = glue("{param} = {val};")  
  ) %>%  
  pull(line) %>%  
  paste(collapse = "\n")  
  
Endpoints_print <- "Weight,  
  Length_struct,  
  Reproduction,  
  Energy,  
  Organic_matter,"  
  
f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)  
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr  
# ./mcsim.DEBcaliAWI DEB_sim_full.in  
  
#####  
df_No_sim <- df_data |>  
  dplyr::select(ID_experiment, No_sim)  
  
df_carac_color <- data.frame(  
  carac = unique(l_carac),
```

```

    color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
              Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
  )
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)

```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between variables
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

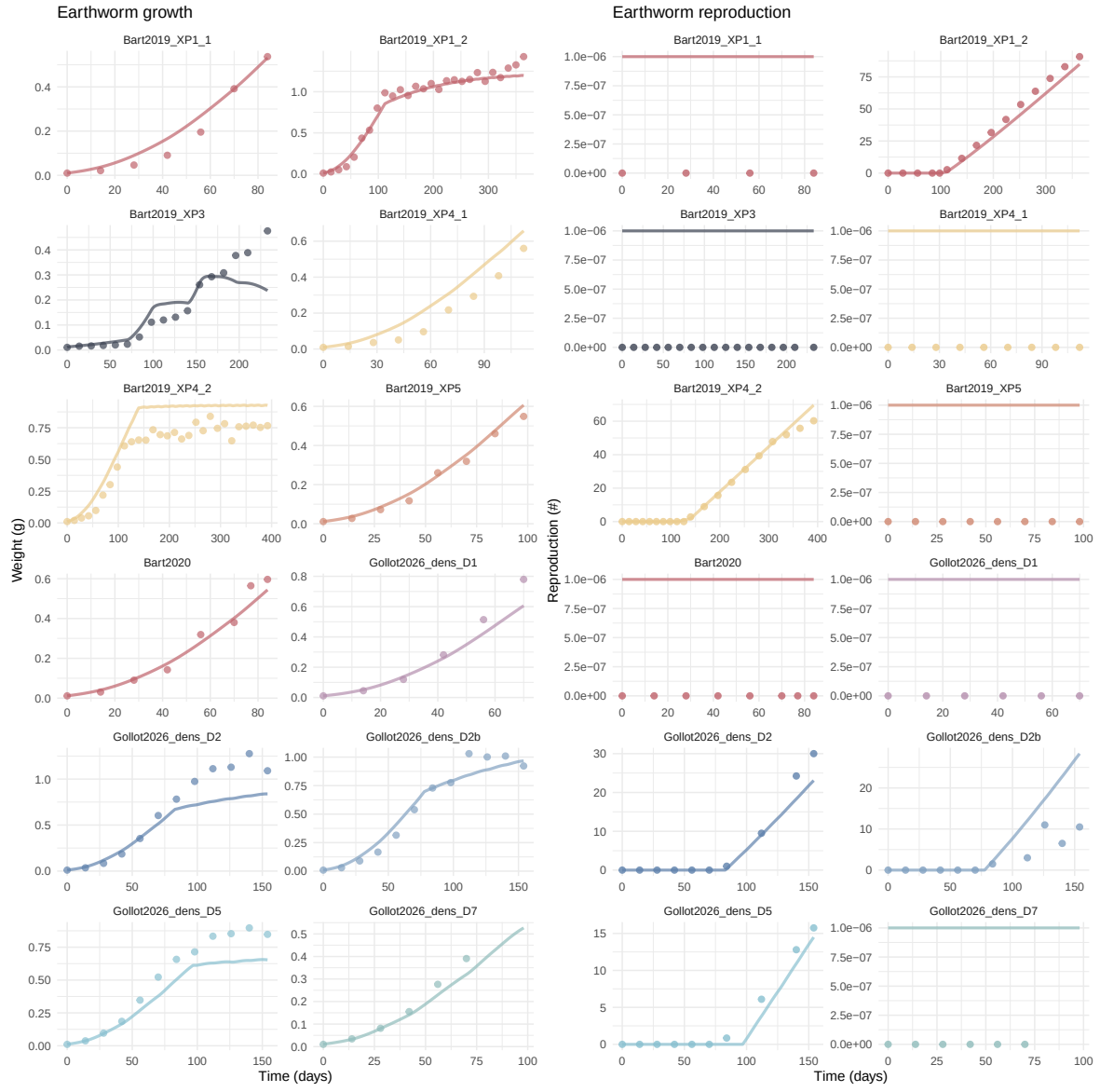
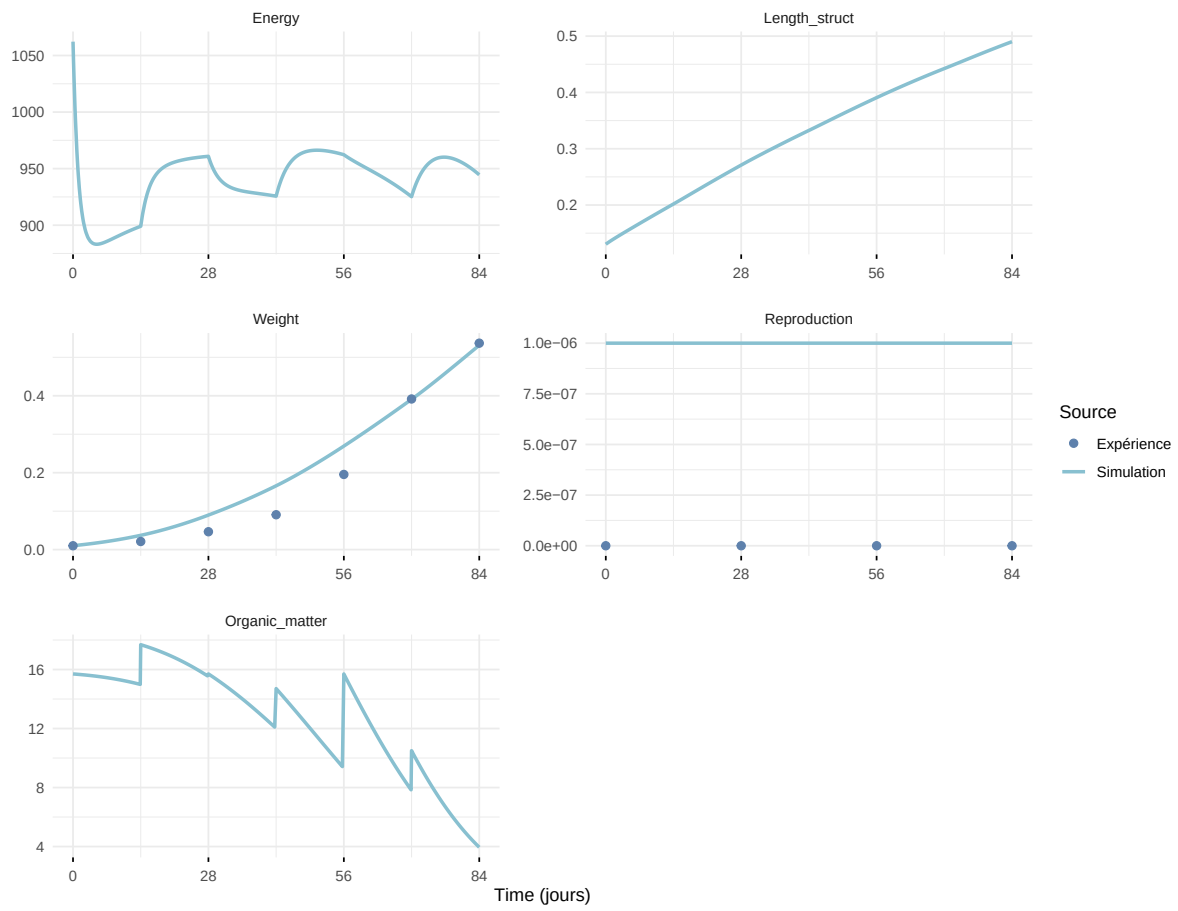
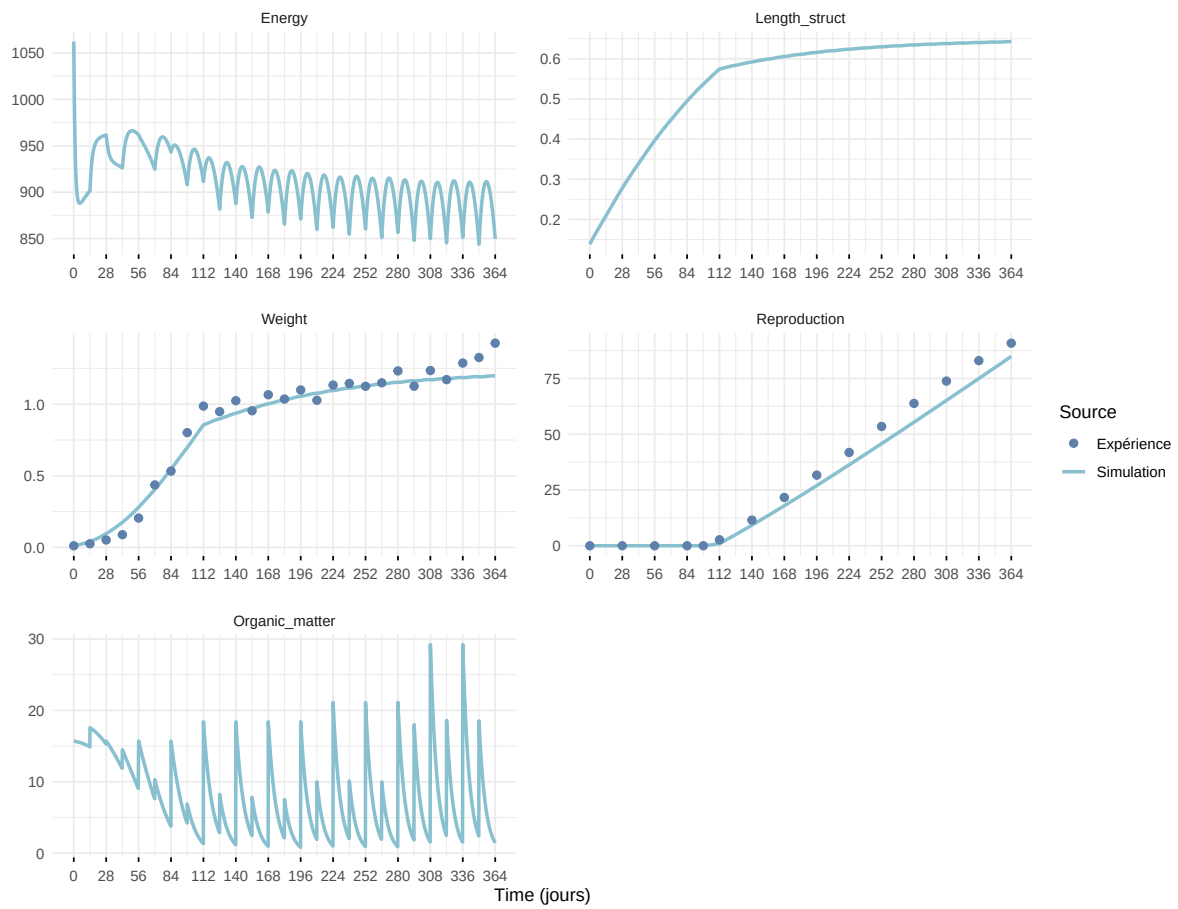


Figure 7: Simulations Weight and Reproduction.

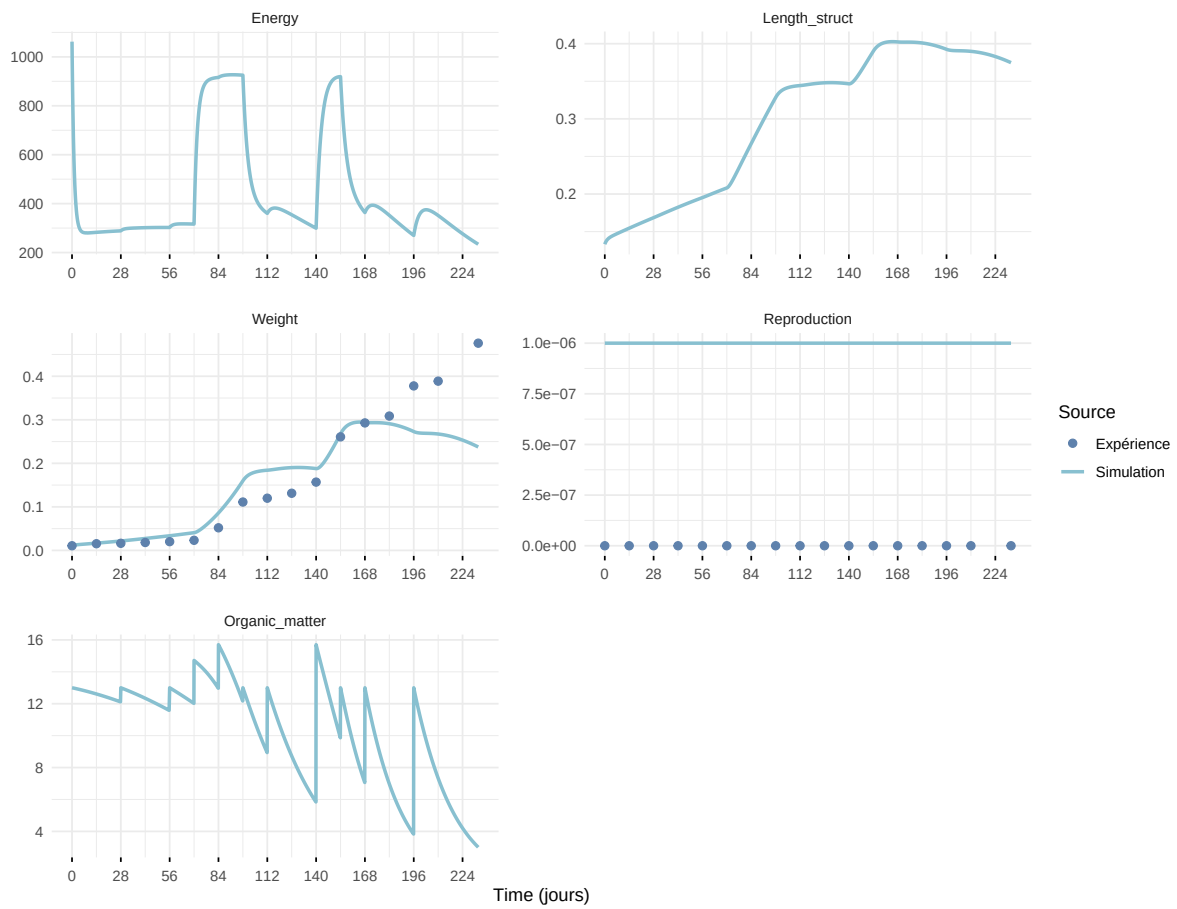
Expérience : Bart2019_XP1_1



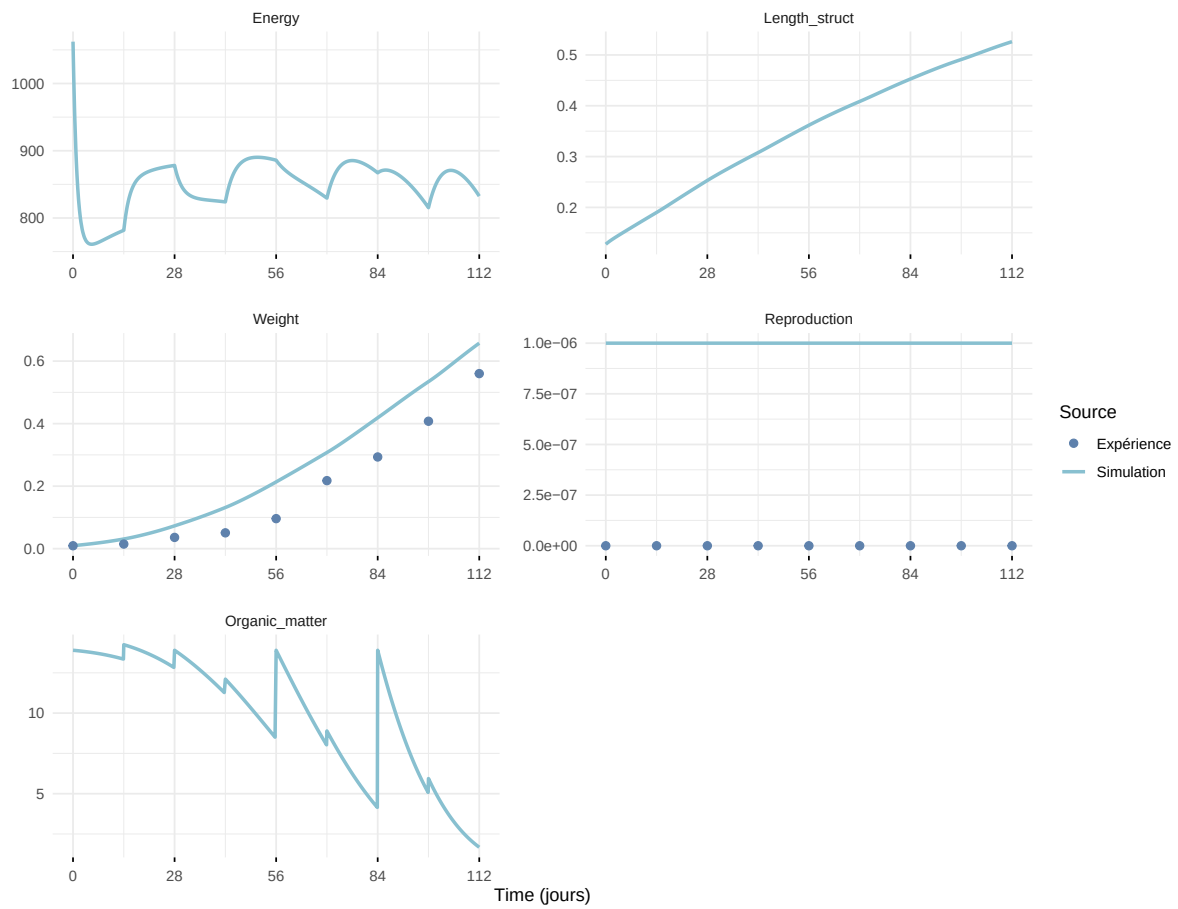
Expérience : Bart2019_XP1_2



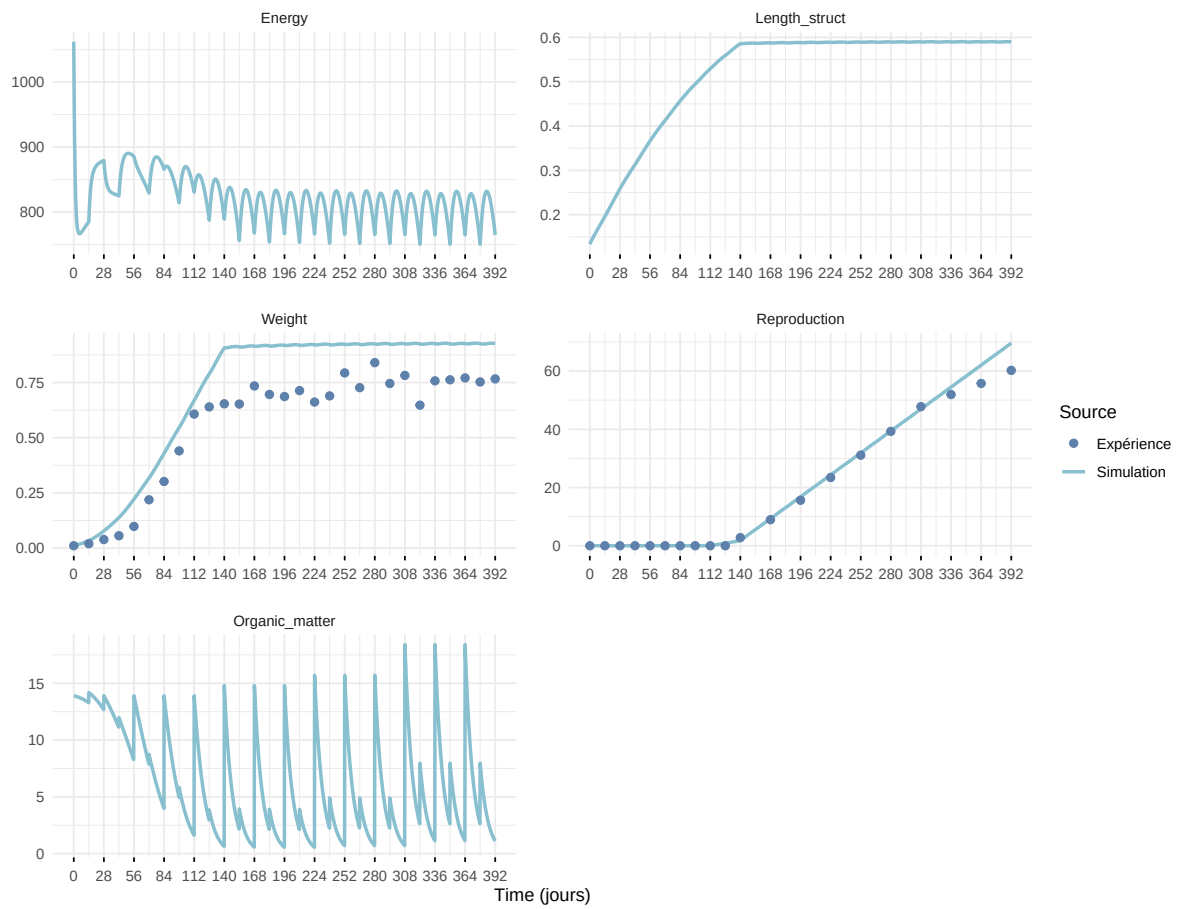
Expérience : Bart2019_XP3



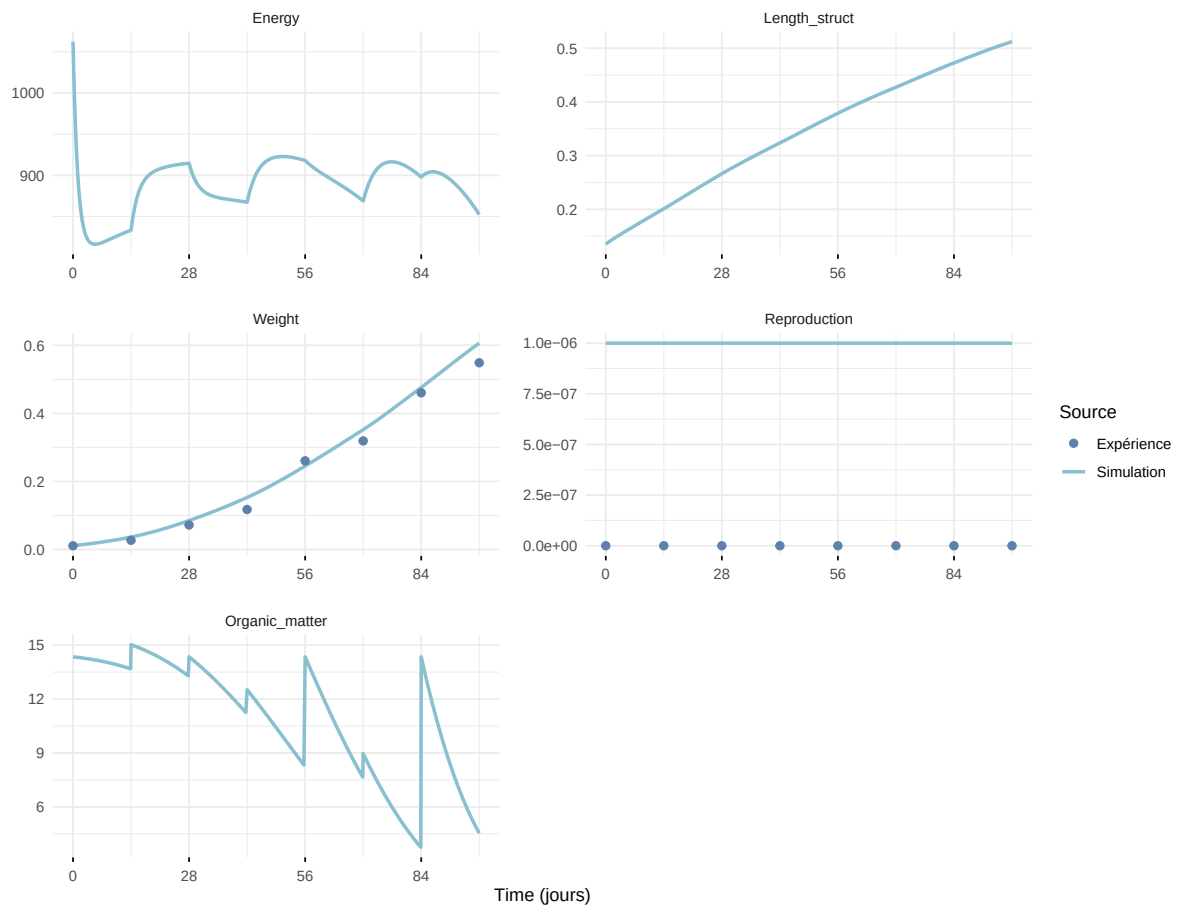
Expérience : Bart2019_XP4_1



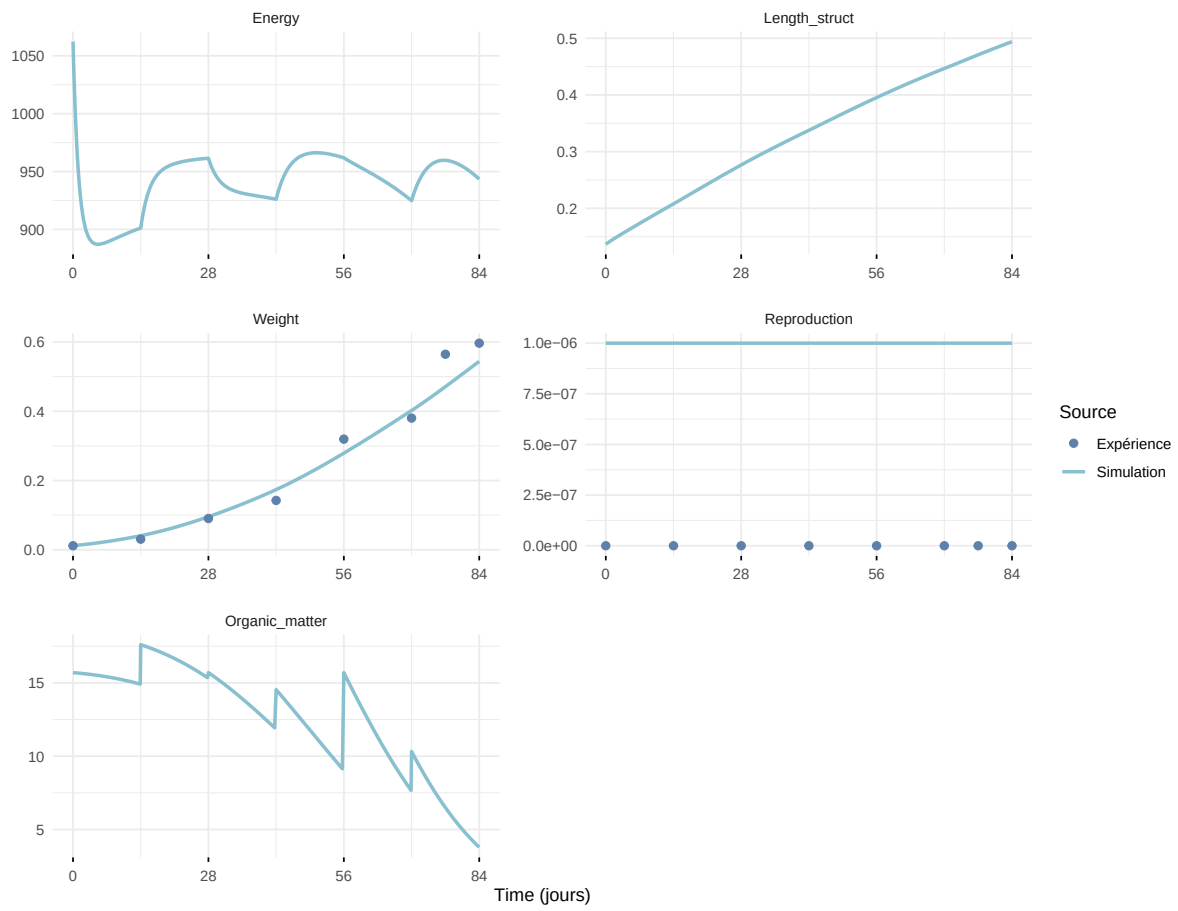
Expérience : Bart2019_XP4_2



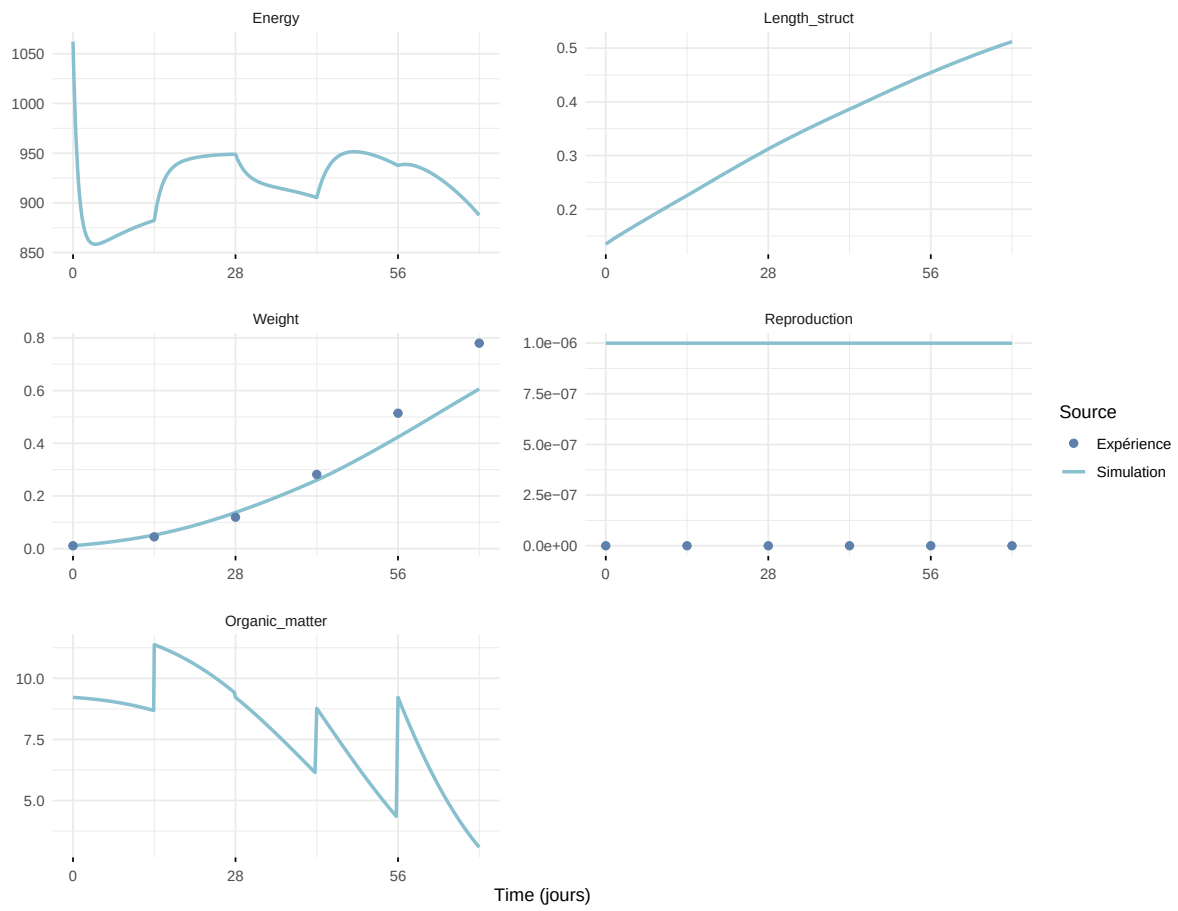
Expérience : Bart2019_XP5



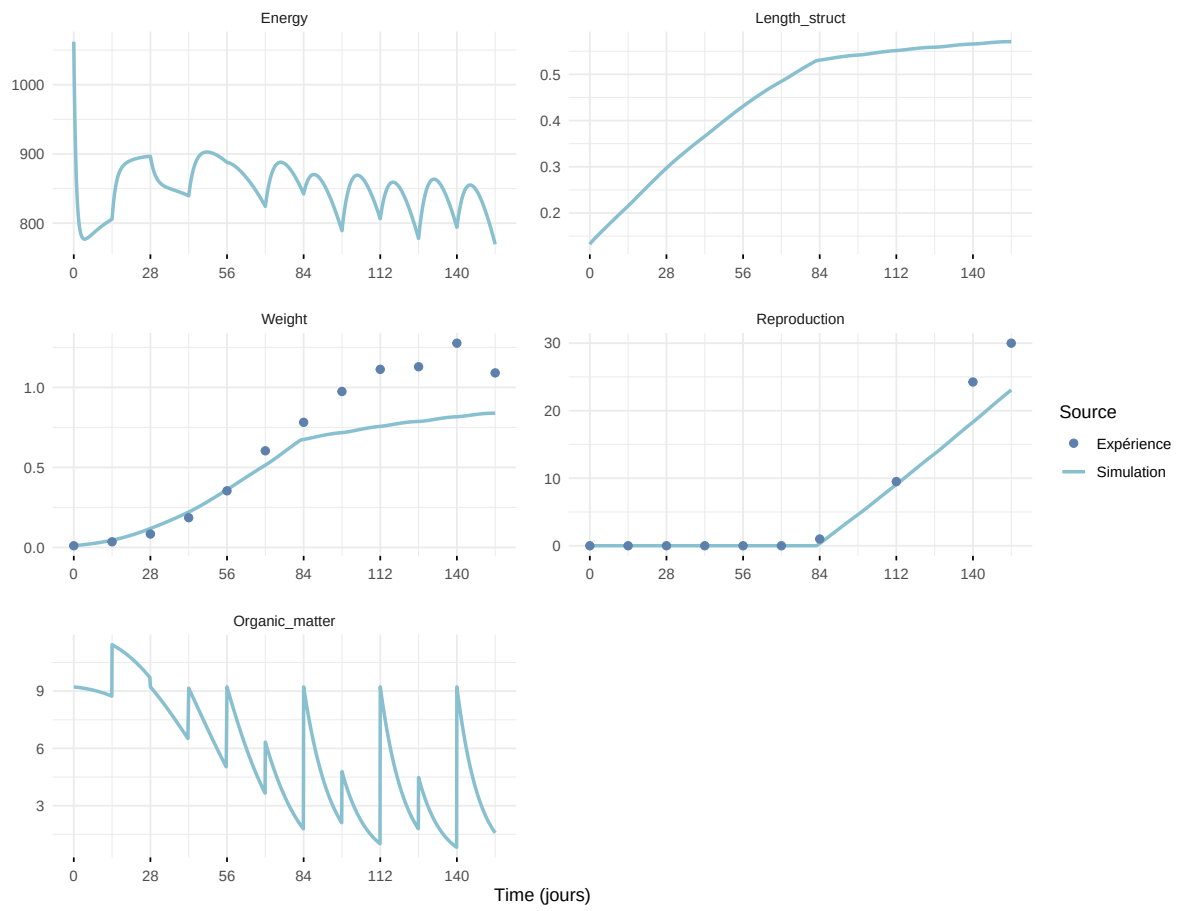
Expérience : Bart2020



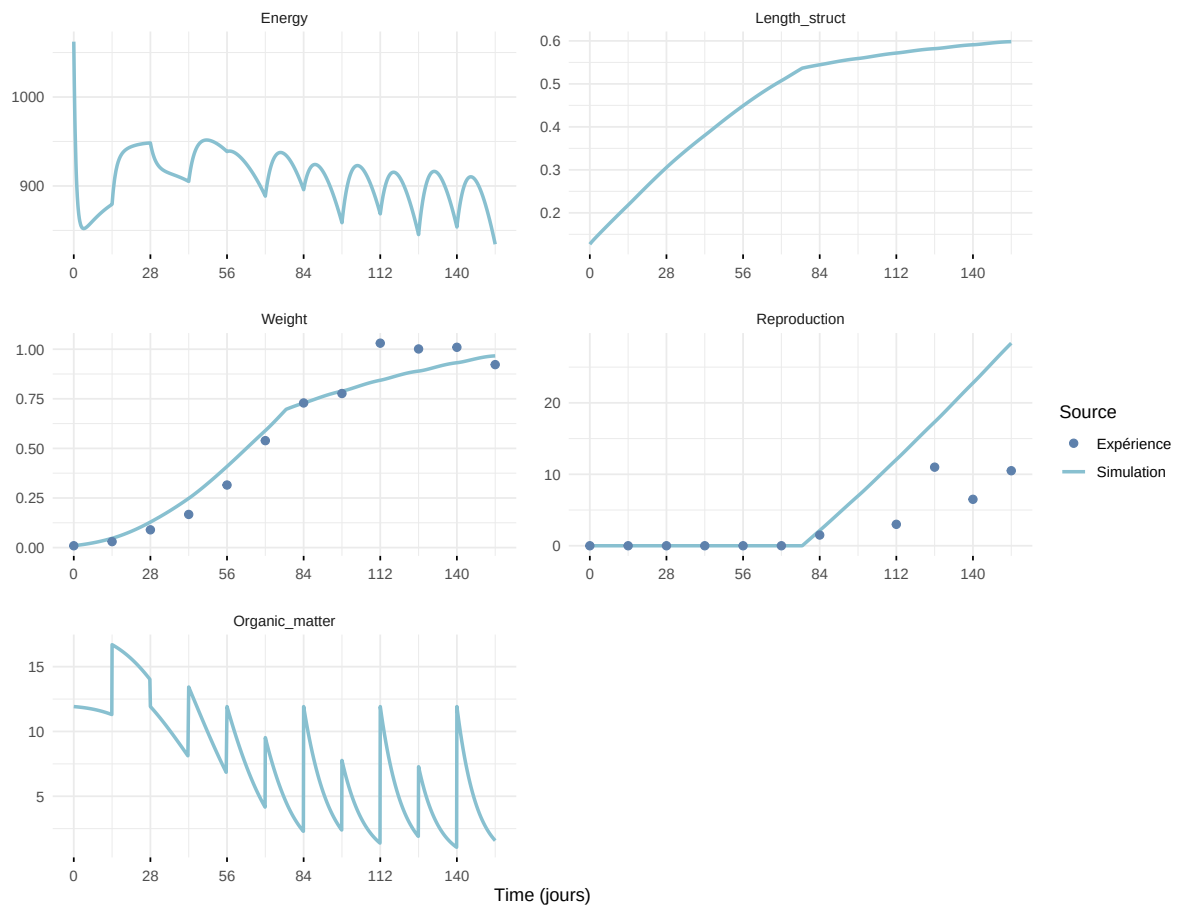
Expérience : Gollot2026_dens_D1



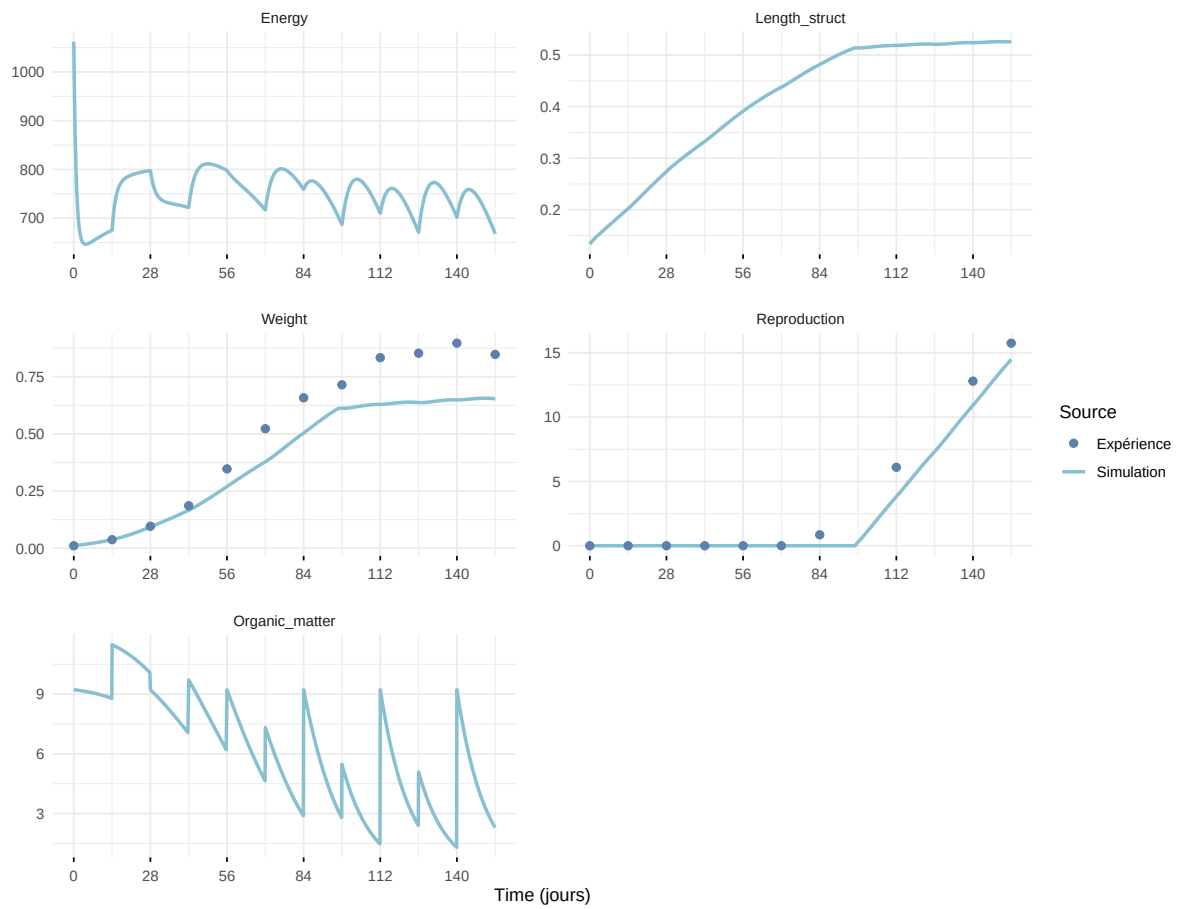
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

